

BKN AI Capital — Independent Technical Due-Diligence Review

Senior software-architect / investor due-diligence perspective • Evaluated as of Sprints 1–13 • Evidence-based

Disclosure: the reviewer contributed to parts of this codebase; scoring is weighted against that bias, not for it. Findings verified against source, tests, configuration, and CI.

OVERALL SCORE

59 / 100

PRODUCTION APPROVAL

NO

Stage: Engineering-beta, product-prototype (pre-seed). High engineering craft wrapped around an unvalidated, non-ML decision engine running on synthetic data by default.

Baseline facts (measured, not claimed)

• **~15,640 LOC backend / ~5,810 LOC frontend.** 44 backend test files, 244 tests, 84% coverage, real Postgres+TimescaleDB integration in CI. **4 frontend test files (all lib utilities) — 0 component, page, or E2E tests** across 10 feature components.

• **Runs on synthetic data by default:** `market_provider = "simulated"`, `market_feed_enabled = false`. The default experience analyzes a random-walk generator, not a market.

• **The "AI Committee" contains no ML and no LLM.** No ML/LLM library appears anywhere in `committee/` or `ai_engine/`. Agents are hand-coded rules over indicators with weighted sums; the base class notes an agent "can later be swapped for an LLM-backed analyst." It is a deterministic expert system labeled "AI."

• **Strategy stats start empty** (`trades = 0`, nothing is `proven`). A backtester exists, but there is no in-repo evidence of a validated edge.

• **Single-process by default:** in-memory `asyncio` event bus, in-process WebSocket gateway (Redis-Streams fan-out exists but is off). Deployment is `docker-compose` only — no k8s/Terraform. Observability is Prometheus + Grafana + `structlog`; no tracing/APM/error-tracking.

Category scores (1–10)

#	Category	Score	Core evidence
1	System Architecture	8/10	Clean modular monolith; provider ports + registry, event bus, DI, domain models. Docked: single-process scaling default.
2	Backend Engineering	8/10	Async SQLAlchemy, Alembic, Pydantic v2, mypy strict, 244 tests, real Timescale in CI. Docked: in-memory shortcuts; scale unproven.
3	Frontend / UI	6/10	Competent Next.js, consistent design system, real API wiring, error boundaries, responsive. Docked hard: near-zero tests; basic charting.
4	User Experience	6/10	Clean dark terminal, proper loading/empty/error states, mobile nav. But shallow: a viewer of recommendations, no tools/alerts/onboarding.
5	Trading Workflow	5/10	Coherent paper engine + journal + analytics, but advisory- and paper-only on simulated data; no live fills or broker reconciliation.
6	AI / Decision Engine	4/10	Explainable rules engine, but no ML/learning, no validated alpha, empty stats. As an AI that makes money: unproven scaffolding.
7	Scalability	4/10	In-memory bus + in-process WS + <code>compose-only</code> + no autoscaling + no load-test evidence.

#	Category	Score	Core evidence
8	Security	7/10	Argon2id, rotating httpOnly refresh with reuse detection, rate-limit + lockout, RBAC, Fernet, audit. Docked: MFA unused, env secrets, no pentest/scanning.
9	Code Quality & Maintainability	8/10	Strict typing, enforced lint/format gates, modular, documented, backend well-tested. Blemish: frontend test hole.
10	Production Readiness	4/10	Fake data default, live pipeline not wired, compose-only, no HA/tracing/SLOs, untested frontend, no compliance layer.
11	Commercial Potential	5/10	Large real market + sensible advisory/explainability angle, but core value (do calls make money?) unvalidated; fierce competition.
12	Overall Engineering Quality	7/10	High craft, thin substance: plumbing is genuinely good; algorithmic/product core is unproven.

Unweighted mean ≈ 6.0 . For an investor, Production Readiness and AI/alpha-validity weigh more, pulling the effective figure slightly lower.

The 10 biggest weaknesses

1. **No demonstrated alpha.** The whole thesis rests on recommendations being good; there is zero in-repo evidence they beat random. Empty strategy stats confirm no track record.
2. **"AI" is deterministic rules.** Positioned as a multi-agent AI committee; it is hand-tuned heuristics. Investors will feel misled if this isn't stated plainly.
3. **Synthetic data by default.** Out of the box it analyzes a random walk. Live Zerodha data is built at the provider layer but not wired end-to-end and needs daily manual token login.
4. **Frontend essentially untested** — 0 component/page/E2E tests. Regressions in the actual product are undefended.
5. **Single-process scaling.** In-memory bus + in-process WS; no proven horizontal path, no load testing.
6. **No production infra.** docker-compose only; no k8s/Terraform/HA/blue-green; secrets in env vars.
7. **No distributed tracing / APM / error tracking.** Counters + logs won't diagnose production incidents.
8. **Daily-token operational fragility** for the one path to real data (Kite requires interactive 2FA every morning).
9. **No regulatory/compliance layer.** Issuing buy/sell recommendations in India (SEBI RA territory) with only a footer disclaimer is a real liability.
10. **Single-tenant assumptions + unused MFA.** Not built for multi-account isolation at scale; MFA field exists but isn't enforced.

Technical debt still present

- Frontend test debt (the largest single gap).
- In-memory event bus / WebSocket as the default transport.
- Money is Decimal at rest but float in the domain layer (paper engine) — fine for paper, unacceptable if it ever touches real money.
- Full Alembic chain can't be dry-run on SQLite (0001 uses Postgres-only CREATE EXTENSION); only Postgres CI validates migrations.
- No dependency/vulnerability scanning or SBOM in CI.
- Recommendation "Committee Vote" is fetched per-row on demand (N scans) — a scaling smell if productized.

What would stop professional use today

- It runs on fake data unless someone wires and babysits a live broker token daily.
- No evidence the signals are profitable — professionals won't risk capital on unvalidated heuristics labeled "AI."
- No live execution and no proven live data pipeline — it's a research/paper tool.
- Compliance gap for issuing financial recommendations.
- No SLOs, HA, tracing, or load testing — not operable under real incident load.

Missing vs TradingView / TrendSpider / QuantConnect / AlgoTest

- **Charting:** no drawing tools, multi-pane/compare, deep indicator set, or chart-based alerting.
- **Strategy authoring:** no user-facing builder/IDE, parameter optimization, walk-forward/Monte-Carlo, or live-vs-backtest reconciliation.
- **Execution & broker breadth:** no live order routing, one broker partially wired, no multi-broker, no options strategy builder/payoff analysis.
- **Data:** no real historical depth/tick archive, fundamentals, cross-asset, or news NLP (currently simulated).
- **Platform:** no alerts/automation, saved scanners, community/social, mobile apps, or marketplace.

• **Validated backtesting at scale** with realistic cost/slippage/borrow modeling.

Valuation stage

Engineering-beta, product-prototype — pre-seed. The software craftsmanship (architecture, backend, code discipline) is at a beta/near-production bar. But the *product* — a system whose recommendations demonstrably make money on live data at scale — is at prototype: it runs on synthetic data, the "AI" is unproven rules, and there is no evidence of edge. Investors buy the product thesis, not the plumbing, so the effective stage is early prototype/MVP with unusually strong engineering foundations. Fundable at pre-seed on build quality and team, **contingent on proving signal quality on real data.**

Overall

Overall score: 59 / 100. A well-architected, disciplined codebase (that part is real and above-average) wrapped around an unvalidated, non-ML decision engine running on synthetic data, with a production-readiness and frontend-testing gap.

Approve for production today? No — not close. Blocking items: (1) defaults to fake data with no validated live pipeline; (2) no evidence the recommendations are profitable; (3) frontend untested; (4) no HA/tracing/load-testing; (5) no compliance framework for financial advice.

Highest-leverage fixes, in order: prove signal quality on real historical data (rigorous, cost-aware backtests with published metrics) → wire and stabilize live data → add frontend/E2E tests and observability → address SEBI-advisory compliance. Fix those and this moves from "impressive engineering demo" to "investable product."

Prepared as an independent technical due-diligence assessment. BKN AI Capital is a decision-support platform; in its current form it adds market analysis and paper trading only and does not execute live trades.